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ON OPTIMAL ALLOCATION OF INDIVISIBLES UNDER UNCERTAINTY

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The optimal allocation of indivisible resources is formalized as a stochastic optimization problem involving discrete decision variables. A general stochastic search procedure is proposed, which develops the concept of the branch-and-bound method. The main idea is to process large collections of possible solutions and to devote more attention to the most promising groups. By gathering more information to reduce the uncertainty and by narrowing the search area, the optimal solution can be found with probability one. Special techniques for calculating stochastic lower and upper bounds are discussed. The results are illustrated by a computational experiment.

The aim of this paper is to develop a stochastic version of the branch-and-bound method for optimization problems involving discrete decision variables and uncertainties. The proposed procedure can be applied in cases when conventional deterministic techniques run into difficulties in calculating exact bounds. Such situations are typical for optimization of stochastic systems with indivisible resources.

To illustrate the complexity encountered, let us recall two classical decision models. At first, consider the following well-known hypothesis testing problem. Suppose that there are two actions $i = 1, 2$ with random outcomes α_i , $i = 1, 2$. The distribution of α_i depends on i , but is unknown. By using random observations of α_i , we want to find the action with the smallest expected outcome $\mathbf{E}\alpha_i$. Obviously, this problem is equivalent to the verification of the inequality: $\mathbf{E}\alpha_1 < \mathbf{E}\alpha_2$. Even this problem with only two alternatives is a nontrivial problem of mathematical statistics.

A more general problem which is often referred to as the automaton learning or the multiarmed bandit problem is the following (see Gittins 1989). Let $\{1, \dots, N\}$ be the set of possible actions of the automaton, and let α_i be the response of the "environment" to action i . Again, the distribution of α_i depends on i but is otherwise unknown. The automaton attempts to improve its behavior (current action) on the basis of the responses to previous actions. In other words, the goal is to find a strategy that generates a sequence of actions $i_0, i_1, \dots$ evolving (in some sense) to the action with the smallest expected outcome

$$\min_{1 \leq i \leq N} \mathbf{E}\alpha_i.$$

Here the number of actions may be larger than two, but it is still assumed that it is possible to test all of them.

Let us now discuss a seemingly similar, but in fact much more difficult example of a discrete stochastic optimization model. The main concern again is the choice among actions with random outcomes, but the set of feasible actions

is given implicitly and their number N may be astronomically large. For example, a feasible action may be associated with a vector $x = (x_1, \dots, x_n)$, satisfying the constraint

$$\sum_{j=1}^n d_j x_j \leq b,$$

where components x_j take on values 0 or 1. The parameters b and d_j , $j = 1, \dots, n$, are assumed to be positive. The outcome of a feasible action x is characterized by a random value

$$\sum_{j=1}^n c_j x_j,$$

where the distribution of random coefficients c_j , $j = 1, \dots, n$ is unknown. As in the automaton learning problem, the main question may be the choice of an action leading to the smallest (largest) expected outcome

$$F(x) = \sum_{i=1}^n \mathbf{E} c_i x_i.$$

This is a stochastic version of the well-known knapsack problem encountered in many applications involving allocation of indivisible resources. A more complicated version of this problem arises when some demands d_j , $j = n_1, \dots, n$, and the resource b are random, too, and we allow the decisions about the values of x_j , $j = n_1, \dots, n$, to be made *after* the values of the random parameters are observed.

In Section 1 we discuss various other applications and models of discrete stochastic optimization. Their common feature is that some indivisible resources of different categories have to be allocated to a number of competing activities, sometimes in successive time periods. An important property of these problems is that the number N of possible solutions (actions) may be very large. Therefore,

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the use of standard hypotheses testing techniques or techniques developed for automaton learning becomes practically impossible, because they are based on sequential observations of outcomes of *all* feasible actions.

There are only few works devoted to solving stochastic discrete programming problems, which are usually concerned with special cases, e.g., Yudin and Tzoy (1974), Dupač and Herkenrath (1982), Stougie (1985), Rinnooy Kan and Stougie (1988), Lageweg et al. (1988), Laporte and Louveaux (1993), Louveaux and van der Vlerk (1993), Schultz et al. (1993), and Van der Vlerk (1995). An alternative approach is to use random search techniques, which aim at finding an acceptable suboptimal solution (Andradottir 1996, Ho and Lau 1997, Tang 1994, and Yan and Mukai 1992).

Generally, the development of solution techniques for stochastic discrete programming problems is in an embryonic state. The purpose of the present paper is to discuss the capabilities and properties of one of the most popular discrete programming methods—the branch-and-bound method—when applied to stochastic discrete programming. The remarkable feature of the branch-and-bound method is that it can combine global search and local search (heuristic) procedures (for calculating bounds) in a natural way.

In the stochastic branch-and-bound method, the whole search area is subdivided into subsets. Random upper and lower bounds are used, which are calculated with an accuracy depending on the size of the set and on previous values of the estimates. The updating procedure consists of calculating more precise bounds for the subsets by using additional observations and/or subdividing some of the subsets. A special role is played by the *record set*, which is the subset having the smallest lower bound. The record set is partitioned at each step of the method. Thus the optimal action (solution) is constructed sequentially without examination of each feasible action. In Section 2 the method is described in detail, and its convergence is proven. The discussion of Section 3 concerns the estimation of stochastic upper and lower bounds. In Section 4 we provide a numerical illustration. Finally, we have a conclusion section.

1. MODELS OF DISCRETE STOCHASTIC OPTIMIZATION

Let us now discuss some stochastic optimization problems that can be approached by using the techniques proposed in the next section.

A rather general model is formulated as follows. Suppose that actions or solutions are characterized by vectors $x = (x_1, \dots, x_n)$ from a finite set $X \subset R^n$. For example, X may be defined as the intersection of some (in particular integer) lattice in R^n with a hypercube in R^n . Often, components of the vector x take on 0–1 values.

Assume that outcomes of an action x can be described by a function $f(x, \theta)$, where $\theta \in \Theta$ and $(\Theta, \Sigma, \mathbf{P})$ is a

probability space. The problem is to find the action x that minimizes the expected outcome

$$F(x) = \mathbf{E} f(x, \theta), \quad (1)$$

among

$$x \in X \cap D, \quad (2)$$

where D is a subset in R^n given, for example, by some inequalities

$$D = \{x \in R^n : g_i(x) \leq 0, i = 1, \dots, m\}.$$

Let us now consider some important applied problems which are later used as illustrations for proposed techniques.

Example 1. Pollution control. In the simplest pollution control problem there are emission sources $i = 1, \dots, m$ and receptor points $j = 1, \dots, n$. For every source i , a finite set $K(i)$ of treatment technologies is available. Each technology $k_i \in K(i)$ has cost c_{ik_i} and is associated with an emission level e_{ik_i} . The emissions are transferred to receptors to produce depositions

$$y_j(k_1, \dots, k_m, \theta) = \sum_{i=1}^m t_{ij}(\theta) e_{ik_i},$$

where $t_{ij}(\theta)$ are some random transfer coefficients. Finally, there are some target levels (ambient norms) of deposition q_j for the receptors $j = 1, \dots, n$. They are used to formulate a penalty cost $\varphi_j(y_j)$ associated with each deposition, e.g.,

$$\varphi_j(y_j) = \max(0, y_j - q_j).$$

The problem is to find the technologies $k_1, \dots, k_m$ so as to minimize the penalty function

$$F(k_1, \dots, k_m) = \mathbf{E} \left\{ \sum_{j=1}^n \varphi_j \left(\sum_{i=1}^m t_{ij}(\theta) e_{ik_i} \right) \right\}$$

subject to the budget constraint

$$\sum_{i=1}^m c_{ik_i} \leq R.$$

Example 2. Facility location. A set $\mathcal{N} = \{1, 2, \dots, n\}$ of potential facility locations and a set of clients $I = \{1, 2, \dots, m\}$ are given. A facility placed at location j costs c_j and has capacity u_j . Clients have random demands $d_i(\theta)$, $i = 1, \dots, m$, and the unit cost to satisfy the demand of client i from facility j is q_{ij} . There is also a shortage cost q_{i0} for each unit of client's i demand, which is not satisfied by any of the facilities. The problem is to choose locations of facilities that minimize the total expected cost.

Defining binary variables

$$x_j = \begin{cases} 1 & \text{if facility is placed at } j, \\ 0 & \text{otherwise,} \end{cases}$$

one can formalize the problem as follows:

$$\min \left[F(x) = \sum_{j=1}^n c_j x_j + \mathbf{E} \varphi(x, \theta) \right], x_j \in \{0, 1\},$$

$$j = 1, \dots, n,$$

where $\varphi(x, \theta)$ is defined as the minimum cost of satisfying the demand. It is the optimal value of the transportation problem

$$\min \sum_{i=1}^m \sum_{j=0}^n q_{ij} y_{ij},$$

$$\sum_{j=0}^n y_{ij} = d_i(\theta), \quad i = 1, \dots, m,$$

$$\sum_{i=1}^m y_{ij} \leq x_j u_j, \quad j = 1, \dots, n,$$

$$y_{ij} \geq 0, \quad i = 1, \dots, m, j = 1, \dots, n,$$

where y_{ij} is the demand of client i served by facility j .

Example 3. Project financing. There are n prospective projects to be implemented. The cost of starting project j is c_j . The projects use resources $i = 1, \dots, m$, initially available in quantities b_i , and the initial demand for resource i by project j is d_{ij} . After the projects are started, additional random quantities of resources $\beta_i(\theta)$ become available, random inputs $\delta_{ij}(\theta)$ of resources necessary to continue the projects, and random incomes $q_j(\theta)$ from successfully completed projects become known. At this stage one has to decide which projects are to be continued. The problem is to find the initial set of projects started so that the expected profit from the whole enterprise is maximized, while the resource constraints at the initial stage and the final stage are satisfied.

Using binary variables

$$x_j = \begin{cases} 1 & \text{if project } j \text{ is started,} \\ 0 & \text{otherwise,} \end{cases}$$

we can define the feasible set as

$$X = \left\{ x \in \{0, 1\}^n : \sum_{j=1}^n d_{ij} x_j \leq b_i, i = 1, \dots, m \right\}.$$

The objective (to be minimized) is given by the function

$$F(x) = \sum_{j=1}^n c_j x_j - \mathbf{E} \varphi(x, \theta), \quad (3)$$

where $\varphi(x, \theta)$ is the actual income from the projects selected to be continued. It is the optimal value of the multi-knapsack problem:

$$\max \sum_{j=1}^n q_j(\theta) y_j,$$

$$\sum_{j=1}^n \delta_{ij}(\theta) y_j \leq b_i + \beta_i(\theta) - \sum_{j=1}^n d_{ij} x_j, \quad i = 1, \dots, m,$$

$$y_j \in \{0, x_j\}, \quad j = 1, \dots, n.$$

Let us observe that even if θ has a finite number of possible values (scenarios) $\theta^1, \dots, \theta^S$, the equivalent determin-

istic formulation of this problem has $n(L + 1)$ binary variables, $m(S + 1)$ knapsack constraints, and mS implication constraints.

Example 4. Expansion of a network. Consider a network with the set of nodes $\mathcal{N}$, the set of arcs $\mathcal{A}$ and with the node-arc incidence matrix M . The arc capacities x_{ij} , $(i, j) \in \mathcal{A}$, have to be chosen from some finite sets X_{ij} with costs $c_{ij}(x_{ij})$. There is a random supply/demand in the network: $d_i(\theta)$, $i \in \mathcal{N}$, and the unit cost of flow on arc (i, j) is q_{ij} . The problem is to invest in arc capacities (select $x_{ij} \in X_{ij}$) so as to minimize the objective function

$$F(x) = \sum_{(i,j) \in \mathcal{A}} c_{ij}(x_{ij}) + \mathbf{E} \varphi(x, \theta).$$

The first part of $F(x)$ represents direct investment costs, while $\varphi(x, \theta)$ represents transportation costs defined as the minimum value of the network flow problem:

$$\min \sum_{(i,j) \in \mathcal{A}} q_{ij} y_{ij},$$

$$My = d(\theta),$$

$$0 \leq y_{ij} \leq x_{ij}, \quad (i, j) \in \mathcal{A}.$$

Example 5. Two-stage network flow problem. Consider a network with the set of nodes $\mathcal{N}$, the set of arcs $\mathcal{A}$ and with the node-arc incidence matrix M . The first-stage decision variables are integer flows (e.g., numbers of flights) x_{ij} for each arc $(i, j) \in \mathcal{A}$. They are nonnegative integers and have to satisfy flow conservation conditions

$$Mx = 0,$$

and some feasibility conditions, e.g.,

$$\sum_{j \in \delta^+(i)} x_{ij} \leq u_i, \quad i \in \mathcal{N},$$

where $\delta^+(i) = \{j : (i, j) \in \mathcal{A}\}$ and u_i is the capacity of node i .

For each pair of nodes $k, l \in \mathcal{N}$ there is a random amount $b^{kl}(\theta)$ of cargo to be delivered from k to l . The unit cost of moving cargo along arc (i, j) is q_{ij} . The problem is to find a feasible first-stage flow x (schedule) such that the total cost

$$F(x) = \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij} + \mathbf{E} \varphi(x, \theta),$$

is minimized. The first part represents direct costs, while $\varphi(x, \theta)$ is the minimum cost to satisfy the demand for cargo shipment once the schedule x is determined. It is the optimal value in the multicommodity network flow problem defined as follows. For each pair (k, l) , we denote the amount of cargo going from k to l through arc (i, j) by y_{ij}^{kl} and the total flow vector by y^{kl} . Let us define the supply/demand vector d^{kl} by

$$d_i^{kl}(\theta) = \begin{cases} b^{kl}(\theta) & \text{if } i = k, \\ -b^{kl}(\theta) & \text{if } i = l, \\ 0 & \text{otherwise.} \end{cases}$$

Then the shipment problem takes on the form

$$\min \sum_{k \in N} \sum_{l \in N} \sum_{(i,j) \in \mathcal{A}} q_{ij} y_{ij}^{kl},$$

$$M y^{kl} = d^{kl}(\theta),$$

$$\sum_{k \in N} \sum_{l \in N} y_{ij}^{kl} \leq x_{ij} v,$$

where v denotes the unit capacity of the first-stage flow (e.g., the maximum payload of an aircraft).

All these examples have common features. There are some indivisible resources to be distributed among many possible activities that make the number of feasible solutions (actions) very large. In addition, exact evaluation of the objective function at any of the feasible points is very difficult. For nontrivial distributions of the uncertain parameters, one can only simulate the random outcomes and calculate some estimates of the objective. It is quite clear that we need a method that would be capable to find a solution without exhaustive examination of all feasible points. In other words, we need a way to direct the search to the regions where promising solutions can be found.

2. THE STOCHASTIC BRANCH-AND-BOUND METHOD

2.1. Outline of the Method

Let us consider the discrete stochastic programming problem

$$\min_{x \in X \cap D} [F(x) = \mathbf{E} f(x, \theta)], \quad (4)$$

where X is a finite set in some solution space $\mathcal{X}$, D is some (possibly infinite) subset of the space $\mathcal{X}$, θ is an elementary event in a probability space $(\Theta, \Sigma, \mathbf{P})$, and $\mathbf{E}$ denotes the mathematical expectation operator.

For example, the n -dimensional Euclidean space R^n may serve as the solution space $\mathcal{X}$, the set D may be given in R^n by some inequalities and the set X may be defined as an intersection of some (in particular integer) lattice in R^n with a bounded box.

In the branch-and-bound method the original set X is iteratively subdivided into subsets X^p generating a partition $\mathcal{P}$ of X . Consequently, the original problem is subdivided into subproblems

$$\min_{x \in X^p \cap D} [F(x) = \mathbf{E} f(x, \theta)], \quad X^p \in \mathcal{P}.$$

Let $F^*(X^p)$ denote the optimal value of this subproblem. Clearly, the optimal value of the whole problem equals

$$F^*(X) = \min_{X^p \in \mathcal{P}} F^*(X^p).$$

The main idea of the stochastic branch and bound method is to iteratively execute three operations:

- (1) partitioning into smaller subsets,
- (2) estimation of the objective within the subsets,
- (3) removal of some subsets.

The procedure continues until some stopping criterion is satisfied, e.g., until promising subsets X^p become singletons. The exhaustive examination of X is reduced by using

statistical estimates of lower and upper bounds of optimal values $F^*(X^p)$.

We make the following assumption.

Assumption 1. *There exist functions $L : 2^X \rightarrow R$ and $U : 2^X \rightarrow R$ such that for each $X^p \subset X$*

$$L(X^p) \leq F^*(X^p) \leq U(X^p),$$

$$U(X^p) = F(x') \quad \text{for some } x' \in X^p,$$

and if X^p is degenerated into a singleton then

$$L(X^p) = F^*(X^p) = U(X^p).$$

We also assume that if $X^p \cap D = \emptyset$ then this case can be identified and, by definition, $L(X^p) = U(X^p) = +\infty$.

The functions L and U are usually obtained from some auxiliary stochastic optimization problems defined on the subsets X^p . In Section 3 we shall discuss some ways to construct such subproblems.

Obviously, the optimal value $F^*(X)$ cannot be achieved on those sets X^p where

$$L(X^p) > \min_{X^q \in \mathcal{P}} U(X^q),$$

so such sets could be deleted from the current partition, if we knew the bounds. However, in stochastic problems the bounds $L(X^p)$ and $U(X^p)$ need to be constructed in a special way (e.g., with the use of the mathematical expectation operator). They can be computed exactly only in some special cases and at the expense of high costs. In general, however, we can only assume that some statistical estimates of $L(X^p)$ and $U(X^p)$ are available.

Assumption 2. *In some probability space $(\Omega, \Sigma, \mathbf{P})$, for each subset $X^p \subset X$, there exist sequences of random estimates $\xi^l(X^p, \omega)$, $l = 1, 2, \dots$, and $\eta^m(X^p, \omega)$, $m = 1, 2, \dots$, $\omega \in \Omega$, such that*

$$\lim_{l \rightarrow \infty} \xi^l(X^p, \omega) = L(X^p) \text{ a.s.},$$

$$\lim_{m \rightarrow \infty} \eta^m(X^p, \omega) = U(X^p) \text{ a.s.}$$

Possible structure of the probability space $(\Omega, \Sigma, \mathbf{P})$ is described in Section 2.3. Let us mention here that if the bounds L and U are defined by some auxiliary stochastic optimization problems with continuous variables, a broad collection of methods can be used to generate estimates satisfying Assumption 2 (see Ermoliev 1976).

2.2. The Algorithm

The key role in the algorithm is played by the record set, i.e., the set that has the least lower bound, and by an approximate solution, which is defined as an element of the subset with the least upper bound. As the record set is partitioned into smaller subsets, new estimates of the objective within the subsets are generated, a new approximate solution is selected, and the iteration continues. Since the bounds are random, the record set is random;

consequently, all objects generated by the method are random. For brevity, we skip the argument ω from the random indices l and m , random partitions $\mathcal{P}$ and random sets.

Step 1. Initialization. Form initial partition $\mathcal{P}_0 = \{X\}$. Calculate the bounds $\xi_0 = \xi^{l_0}(X)$ and $\eta_0 = \eta^{m_0}(X)$. Set $k = 0$.

Step 2. Partitioning. Select the record subset

$$Y^k = \operatorname{argmin}\{\xi_k(X^P) : X^P \in \mathcal{P}_k\},$$

and an approximate solution

$$x^k \in X^k = \operatorname{argmin}\{\eta_k(X^P) : X^P \in \mathcal{P}_k\}.$$

If the record subset is a singleton, then set $\mathcal{P}'_k = \mathcal{P}_k$ and go to Bound Estimation. Otherwise construct a partition of the record set, $\mathcal{P}''_k(Y^k) = \{Y^k_i, i = 1, 2, \dots, n_k\}$. Define new full partition

$$\mathcal{P}'_k = (\mathcal{P}_k \setminus Y^k) \cup \mathcal{P}''_k(Y^k).$$

Elements of $\mathcal{P}'_k$ will also be denoted by X^P .

Step 3. Bound estimation. For all subsets $X^P \in \mathcal{P}'_k$ select some estimates $\xi_{k+1}(X^P) = \xi^{l_{k+1}(X^P)}(X^P)$ and $\eta_{k+1}(X^P) = \eta^{m_{k+1}(X^P)}(X^P)$ for $L(X^P)$ and $U(X^P)$, respectively.

Step 4. Deletion. Clean partition $\mathcal{P}'_k$ of infeasible subsets, defining

$$\mathcal{P}_{k+1} = \mathcal{P}'_k \setminus \{X^P : X^P \cap D = \emptyset\}.$$

Set $k := k + 1$ and go to Partitioning.

If the estimates are exact, i.e., if $\xi_{k+1}(X^P) = L(X^P)$ and $\eta_{k+1}(X^P) = U(X^P)$, then at the Deletion step we can also delete all sets X^P for which $\xi_{k+1}(X^P) > \eta_{k+1}(X^P)$.

2.3. Convergence

In the deterministic case one need not prove convergence of the branch-and-bound method, owing to the finite number of possible solutions. On the contrary, convergence in the stochastic case requires some validation, because of the probabilistic character of bound estimates. The situation is similar to the hypothesis testing problem with two possible decisions and random outcomes, but much more involved. For example, due to random errors, a subset containing the global solution may not be the record set and may remain unpartitioned. Therefore, if we stop the algorithm after a finite number of iterations, the probability of an error and the size of the error have to be estimated.

First of all, let us construct a probabilistic model of the basic algorithm. Assume that partitioning is done by some deterministic rule $\mathcal{P}''$: for every subset $Y \subset X$, $\mathcal{P}''(Y)$ is a collection of disjoint subsets Y_j of Y such that $\cup_j Y_j = Y$. Consider a deterministic tree $T(X)$ obtained from the initial set X by sequential application of the rule $\mathcal{P}''$ to all sets arising in this process, until they become singletons. The set X is the root node. At level 1 there are nodes corresponding to the subsets in $\mathcal{P}''(X)$. Level 2 contains all sets

of $\mathcal{P}''(Y)$ for all $Y \in \mathcal{P}''(X)$, etc. For each set $X' \in T(X)$, we denote by $k(X')$ the location depth of X' in $T(X)$.

Suppose that for each set $X' \in T(X)$ there exists a probability space $(\Omega_{X'}, \Sigma_{X'}, P_{X'})$ such that for all subsets $X'' \in \mathcal{P}''(X')$ there are sequences of random estimates

$$\xi^l(X'', \omega'), \quad \omega' \in \Omega_{X'}, \quad l = 1, 2, \dots,$$

for $L(X'')$ and

$$\eta^m(X'', \omega'), \quad \omega' \in \Omega_{X'}, \quad m = 1, 2, \dots,$$

for $U(X'')$. Denote by

$$(\Omega_k, \Sigma_k, \mathbf{P}_k) = \prod_{\{X' \in T(X) : k(X')=k\}} (\Omega_{X'}, \Sigma_{X'}, \mathbf{P}_{X'}),$$

the product of probability spaces $(\Omega_{X'}, \Sigma_{X'}, \mathbf{P}_{X'})$ over all X' that may arise at iteration k . By construction, the algorithm performs no more than N partitions, where N is a number of elements of X . Let us consider the product of probability spaces:

$$(\Omega, \Sigma, \mathbf{P}) = (\Omega_0 \times \dots \times \Omega_N, \Sigma_0 \times \dots \times \Sigma_N, \mathbf{P}_0 \times \dots \times \mathbf{P}_N),$$

and denote $\omega = (\omega_0, \dots, \omega_N) \in \Omega$. We shall consider all random objects produced by the algorithm as defined on this general probability space.

Let us denote by X^* the solution set of (4) and by F^* the optimal value of the objective.

Theorem 1. Assume that the indices $l_k(X^P)$ and $m_k(X^P)$ are chosen in such a way that, if a subset $X' \in \mathcal{P}_k$ for infinitely many k , then

$$\lim_{k \rightarrow \infty} l_k(X') = \lim_{k \rightarrow \infty} m_k(X') = \infty \text{ a.s.} \quad (5)$$

Then with probability one there exists an iteration number k_0 such that for all $k \leq k_0$

- (i) the record sets Y^k are singletons and $Y^k \subset X^*$;
- (ii) the approximate solutions $x^k \in X^*$.

Proof. Owing to the finite number of elements in X , there can be only a finite number of iterations with partitioning. Therefore, there exists k_1 such that for $k \geq k_1$ all record sets are singletons and the partition remains unchanged. Denote it $\mathcal{P}_\infty$. Observe, that there is only a finite number of possible partitions $\mathcal{P}_\infty$. Let us define *recurrent* record sets as those, which are record sets for infinitely many k . Because the number of record sets is finite, there exists k_2 such that for $k \geq k_2$ all record sets are recurrent. For each recurrent record set $Y \in \mathcal{P}_\infty$, at infinitely many k one has

$$\xi^{l_k(Y)}(Y) \leq \xi^{l_k(X^P)}(X^P),$$

for all subsets X^P that remain in the partition $\mathcal{P}_\infty$, $k \geq k_2$. Passing to the limit in the last inequality, we obtain

$$F(Y) \leq L(X^P) \text{ a.s.,}$$

for all subsets $X^P \in \mathcal{P}_\infty$, which completes the proof of assertion (i).

Consider now the approximate solutions $x^k \in X^k$. By definition

$$\eta^{m_k(X^k)}(X^k) \leq \eta^{m_k(Y^k)}(Y^k).$$

By assumption, and due to the finiteness of X , $m_k(Y^k) \geq 3$ a.s., so $\eta^{m_k(Y^k)}(Y^k) \geq F^*$ a.s. by the first part of the proof. Thus

$$\limsup_{k \rightarrow \infty} \eta^{m_k(X^k)}(X^k) \leq F^* \text{ a.s.}$$

Let a set $X' \in \mathcal{P}_\infty$ be the set X^k with the least upper bound for infinitely many k , and let the point $x' \in X'$ be chosen infinitely often as the point x^k . By finiteness of the partition $\mathcal{P}_\infty$ and by finiteness of the sets, there exists k_3 such that for all $k \geq k_3$ each X^k and each x^k is recurrent in the above sense. Then

$$F(x') = \lim_{k \rightarrow \infty} \eta^{m_k(X')}(X') \leq F^* \text{ a.s.,}$$

i.e., $x' \in X^*$ a.s.. Taking $k_0 = \max\{k_1, k_2, k_3\}$ completes the proof. $\square$

Let us now discuss some issues concerning possible implementation of the conceptual method discussed above.

Stochastic lower and upper bounds for the subproblems can be calculated by making some experiments (observations) on the subproblems. In the next section we describe some general rules for calculating the bounds. In any case, however, with no loss of generality one can assume that the numbers l and m in Assumption 2 correspond to the numbers of observations. The assumptions of the theorem can be satisfied by making new observations for each subset at infinitely many iterations (not quitting observations for any of the subsets). This is a major difference between the stochastic method and the deterministic branch and bound method: we do not delete subsets X' for which

$$\xi_k(X') > \eta_k(X^k).$$

Nevertheless, there is much freedom in specifying the amount of attention devoted to each set, as the following lemma shows.

Lemma 1. *Let the observations for the subsets be made according to the rules*

$$l_k(X^p, \omega) = l_{k-1}(X^p, \omega) + I_k(X^p, \omega),$$

$$m_k(X^p, \omega) = m_{k-1}(X^p, \omega) + J_k(X^p, \omega),$$

with $l_{k-1}(X^p, \omega) = 0$ and $m_{k-1}(X^p, \omega) = 0$ for newly created sets $X^p \in \mathcal{P}_\infty(Y^k)$. Assume that for every $X^p \in \mathcal{P}_\infty$ with probability one

$$\sum_{k=\nu(X^p)}^{\infty} \mathbf{P}\{I_k(X^p, \omega) > 0 | X^p, l_{k-1}(X^p, \omega)\} = \infty,$$

and

$$\sum_{k=\nu(X^p)}^{\infty} \mathbf{P}\{J_k(X^p, \omega) > 0 | X^p, m_{k-1}(X^p, \omega)\} = \infty,$$

where $\nu(X^p)$ is the iteration number at which X^p was created. Then with probability one, for every $X^p \in \mathcal{P}_\infty$ condition (5) is satisfied.

Proof. The result follows immediately from the Borel-Cantelli theorem. $\square$

It seems reasonable to make the frequencies of observations dependent on the estimated quality of the subsets. For example, at each iteration one may allocate a fixed number of observations to the record set (or its newly created subsets, if it was partitioned) and another number of observations to all remaining sets. The choice of the particular nonrecord sets observed at the current iteration can be random; for example, with equal probabilities. In this way the assumptions of the theorem are satisfied, but non-prospective subsets are observed with a low frequency. Another possibility would be to assign some unequal probabilities $\pi(X^p)$ of observations to the subsets. The probabilities may be functions of the current estimates. In particular, the idea of observing sets with the smallest lower bounds, introduced for the multiarmed bandit problem in Lai (1987), may prove successful here because it resolves in a natural way the conflict between estimation and optimization.

Another important implementational and theoretical issue is the stopping criterion. Clearly, because of the stochastic nature of the bounds, a solution obtained after a finite number of observations is, in general, an approximation. Only some probabilistic statements can be made about its accuracy.

Lemma 2. *Assume that the algorithm stops at iteration s and that we can build for all $X^p \in \mathcal{P}_s$ confidence intervals $[\underline{\xi}(X^p), \infty)$ for $L(X^p)$ and a confidence interval $(-\infty, \bar{\eta}(x^s)]$ for $F(x^s)$, $x^s \in X^s$, such that*

$$\mathbf{P}\{\forall (X^p \in \mathcal{P}_s) \underline{\xi}(X^p) \leq L(X^p) \text{ and } F(x^s) \leq \bar{\eta}(x^s)\} \geq 1 - \delta.$$

Then, with probability at least $1 - \delta$,

$$F(x^s) - F^* \leq \bar{\eta}(x^s) - \min_{X^p \in \mathcal{P}_s} \underline{\xi}(X^p). \quad (6)$$

Proof. With probability not smaller than $1 - \delta$,

$$F^* \geq \min_{X^p \in \mathcal{P}_s} L(X^p) \geq \min_{X^p \in \mathcal{P}_s} \underline{\xi}(X^p)$$

and $F(x^s) \leq \bar{\eta}(x^s)$. Combining these two inequalities we obtain the required result. $\square$

It is clear from the error estimate (6) that the quality of the approximate solution x^s can be improved by making $\bar{\eta}$ small (that is the motivation for the choice of x^s) and by moving up lower bounds of the confidence intervals for $L(X^p)$. It suggests that more observations should be devoted to nonrecord subsets that have small $\underline{\xi}(X^p)$.

It should be noted, though, that the confidence intervals cannot be constructed in a straightforward way. At the iteration s , the numbers of observations devoted to particular subsets of the final partition are random. Indeed, such a number for a set X^p depends on the iteration $\nu(X^p)$ at which X^p was created. It also may depend, if nontrivial rules for generating numbers of observations are used, on

the outcomes of observations, e.g., on the number of times X^P was a record set.

The basic procedure allows a number of modifications. Sets of reasonably good suboptimal solutions are often rather rich and it may be sufficient to find one of them by stopping the method relatively early. It is possible to introduce simple (random) rules for cleaning partitions at Step 3 that allow to reach a recurrent partitioning Π_∞ (different in general from $\mathcal{P}_\infty$) fairly quickly. The recurrent singleton record set from Π_∞ may be selected as an approximate solution of the problem. For example, the rule may be to keep at each step the number of subsets in a partitioning within a given bound by deleting them with equal probability. In general, such a rule must guarantee that Π_∞ contains a recurrent record set from $\mathcal{P}_\infty$ with a positive probability. In order to preserve convergence with probability 1, we need to introduce a regenerative feature of the modified procedure. The idea is to restart the calculations after reaching a recurrent record singleton from Π_∞ . We employ this idea in Section 4.

3. ESTIMATION OF LOWER AND UPPER BOUNDS

The main question which remains to be answered is the estimation of the lower and upper bounds. In this section we shall discuss two general ideas:

(1) interchange of minimization and mathematical expectation operators, and

(2) dual estimates.

They will be used to construct other discrete or continuous optimization problems which have their optimal values below the optimal value of the original problem. Clearly, some well-known deterministic methods for generating bounds (such as relaxation of the integrality condition) may be used together with the ideas discussed here.

3.1. Interchange of Minimization and Mathematical Expectation Operators

Consider a discrete stochastic optimization problem without additional general constraints:

$$\min_{x \in X} \mathbf{E} f(x, \theta).$$

The following estimate is true:

$$F^*(X) = \min_{x \in X} \mathbf{E} f(x, \theta) \geq \mathbf{E} \min_{x \in X} f(x, \theta) = \mathbf{E} f(x^*(\theta), \theta),$$

where

$$x^*(\theta) = \arg \min_{x \in X} f(x, \theta).$$

Thus the quantity

$$L(X) = \mathbf{E} f(x^*(\theta), \theta),$$

provides a lower estimate for the optimal value $F^*(X)$. In many cases, for a fixed θ , it is easy to find $x^*(\theta)$.

The lower bound $L(X)$ can be improved by noting that for every n -dimensional random vector $\lambda(\theta)$ such that $\mathbf{E}\lambda(\theta) = 0$ the following relations hold:

$$\begin{aligned} F^*(X) &= \min_{x \in X} \mathbf{E} f(x, \theta) = \min_{x \in X} \mathbf{E} [f(x, \theta) + \lambda(\theta)^T x] \\ &\geq \mathbf{E} \min_{x \in X} [f(x, \theta) + \lambda(\theta)^T x]. \end{aligned} \quad (7)$$

This yields a valid lower bound

$$L_\lambda(X) = \mathbf{E} [f(x_\lambda^*(\theta), \theta) + \lambda(\theta)^T x_\lambda^*(\theta)],$$

where

$$x_\lambda^*(\theta) = \arg \min_{x \in X} [f(x, \theta) + \lambda(\theta)^T x].$$

Of course, the quality of this bound depends on the choice of the function $\lambda(\theta)$, which may be interpreted as the value of information. We shall illustrate its use in Section 4.

A stochastic estimate of $L_\lambda(X)$ can be calculated in an acceptable time by means of a Monte Carlo simulation technique:

$$L_n = \frac{1}{n} \sum_{i=1}^n (f(x_\lambda^*(\theta_i), \theta_i) + \lambda(\theta_i)^T x_\lambda^*(\theta_i)),$$

where θ_i , $i = 1, \dots, n$, are i.i.d. random variables with distribution $\mathbf{P}$.

As an upper bound $U(X)$ for the optimal value $F^*(X)$ the value of the objective function at some feasible point $x' \in X$ can be taken:

$$U(X) = F(x') = \mathbf{E} f(x', \theta).$$

It is important to choose the point x' in such a way that $F(x')$ is as small as possible. Such points can be found by any (heuristic) local stochastic discrete optimization method like, for instance, the stochastic approximation (over integer lattice) method (Dupač and Herkenrath 1982) or a descent direction method (Sergienko 1988).

The same idea of interchanging minimization and mathematical expectation operators can be applied to two-stage stochastic programming problems:

$$\min_{x \in X} [F(x) = f_1(x) + \mathbf{E} \min_{y \in Y(x, \theta)} f_2(x, y, \theta)],$$

where $x \in X$, $\theta \in \Theta$, $y \in Y(x, \theta)$, X , and Θ are some sets; $Y(x, \theta)$ is a multivalued mapping; $(\Theta, \Sigma, \mathbf{P})$ is a probability space; x is a deterministic first-stage solution; $y(\cdot)$ is a random second-stage solution (correction); and $f_1(x)$ and $f_2(x, y, \theta)$ are performance measures related to the first and the second stage, respectively. Let $F^*(X)$ be the optimal value. Then for every $\lambda(\theta)$ such that $\mathbf{E}\lambda(\theta) = 0$ the following estimate holds:

$$\begin{aligned} F^*(X) &\geq \mathbf{E} \left\{ \min_{x \in X, y \in Y(x, \theta)} [f_1(x) \right. \\ &\quad \left. + \lambda(\theta)^T x + f_2(x, y, \theta)] \right\} = L_\lambda(X). \end{aligned} \quad (8)$$

We assume here, of course, that the expectation operation is well-defined. Internal minimization problems, under fixed θ , can often be solved quickly, as the following example shows.

Example 6. Project financing (continued). Let us consider the project financing problem of Example 3 and a subproblem

$$\min_{x \in X \cap X'} F(x),$$

with

$$X' = \{x \in \{0, 1\}^n : l_j \leq x_j \leq u_j, j = 1, \dots, n\},$$

where $l_j, u_j \in \{0, 1\}, j = 1, \dots, n$. Denote

$$Y(x, \theta) = \left\{ y \in \{0, 1\}^n : y \leq x, \right.$$

$$\left. \begin{aligned} \sum_{j=1}^n d_{ij}x_j + \sum_{j=1}^n \delta_{ij}(\theta)y_j \\ \leq b_i + \beta_i(\theta), i = 1, \dots, m \end{aligned} \right\}.$$

$$Z(X', \theta) = \left\{ (x, y) \in X' \times \{0, 1\}^n : \right.$$

$$\left. \begin{aligned} \sum_{j=1}^n d_{ij}x_j \leq b_i, i = 1, \dots, m, y \leq x, \\ \sum_{j=1}^n d_{ij}x_j + \sum_{j=1}^n \delta_{ij}(\theta)y_j \\ \leq b_i + \beta_i(\theta), i = 1, \dots, m \end{aligned} \right\}.$$

The optimum value of the subproblem can be estimated from below as follows:

$$\begin{aligned} \min_{x \in X \cap X'} \mathbf{E} \min_{y \in Y(x, \theta)} \sum_{j=1}^n (c_j x_j - q_j(\theta) y_j) \\ \geq \mathbf{E} \min_{(x, y) \in Z(X', \theta)} \sum_{j=1}^n (c_j x_j - q_j(\theta) y_j). \end{aligned} \quad (9)$$

If $c \geq 0$ and $d \geq 0$, minimization with respect to x on the right-hand side of (9) can be carried out analytically: $\hat{x}_j(\theta) = \max\{l_j, \hat{y}_j(\theta)\}$, while $\hat{y}(\theta)$ is the solution of the following multiknapsack problem:

$$\min \sum_{j=1}^n (c_j - q_j(\theta)) y_j, \quad (10)$$

$$\sum_{j=1}^n (1 - l_j) d_{ij} y_j \leq b_i - \sum_{j=1}^n l_j d_{ij}, \quad i = 1, \dots, m, \quad (11)$$

$$\sum_{j=1}^n ((1 - l_j) d_{ij} + \delta_{ij}(\theta)) y_j \leq b_i + \beta_i(\theta) - \sum_{j=1}^n l_j d_{ij}, \quad (12)$$

$$i = 1, \dots, m,$$

$$y_j \in \{0, u_j\}, \quad j = 1, \dots, n. \quad (13)$$

Note that (11) guarantees the feasibility of $\hat{x}$.

Consequently, the lower bound is obtained by postponing the decision to start the projects until the uncertain data become known. The multiknapsack problem (10)–(13) is much simpler than the original problem. Moreover,

any lower bound for this problem may serve our purposes. For example, one can relax the integrality conditions in (13).

By (7) and (8), for any random vector $\lambda(\theta)$ such that $\mathbf{E}\lambda(\theta) = 0$ and $\mathbf{P}\{c + \lambda(\theta) \geq 0\} = 1$, we can modify (10)–(13) as follows:

$$\min \sum_{j=1}^n (c_j + \lambda_j(\theta) - q_j(\theta)) y_j, \quad (14)$$

subject to (11)–(13). Clearly, the choice of the appropriate modification $\lambda(\theta)$ is crucial here; Section 4 shows how this can be done in the special case of one resource.

If the distribution of θ is discrete with finitely many scenarios $\theta^1, \dots, \theta^S$, we can calculate the expectation of the optimal value in (9) exactly, by solving the multiknapsack problem (14) or its relaxation S times.

As the upper bound $U(X \cap X')$ we can take the objective value

$$F(x') = \sum_{j=1}^n c_j x'_j - \mathbf{E}\varphi(x', \theta),$$

at the solution $x' \in X \cap X'$ of the following approximation of the original problem:

$$\min \sum_{j=1}^n (c_j - \mathbf{E}q_j(\theta)) x_j,$$

$$\sum_{j=1}^n (d_{ij} + \mathbf{E}\delta_{ij}(\theta)) x_j$$

$$\leq b_i + \mathbf{E}\beta_i(\theta), \quad i = 1, \dots, n, \quad x \in X \cap X'.$$

Optimal values of the second stage multiknapsack problem $\varphi(x', \theta)$ for randomly generated realizations of θ may then be used to construct random upper bounds.

An important matter is the reduction of variance in these kinds of statistical estimates. This is discussed in Dantzig and Glynn (1990) and Infanger (1992).

3.2. Dual Estimates in One-stage Stochastic Problems

Dual estimates in combination with nonsmooth optimization methods are widely used in deterministic discrete programming (see Minoux 1989, Nemhauser and Wolsey 1988, and Shor 1985). Let us discuss properties of dual estimates when they are applied to stochastic discrete programming problems.

Consider a general stochastic programming problem:

$$\min[F(x) = \mathbf{E}f(x, \theta)], \quad (15)$$

subject to

$$G_i(x) = \mathbf{E}g_i(x, \theta) \leq 0, \quad i = 1, \dots, m; \quad (16)$$

$$x \in X; \quad (17)$$

where X is some compact (in particular discrete) set, $\theta \in \Theta$, where $(\Theta, \Sigma, \mathbf{P})$ is a probability space, and $\mathbf{E}$ is a mathematical expectation operator. Some nonlinear inequality constraints may be deterministic. Denote by F^*

the optimal value of the problem and define the Lagrangian function

$$\begin{aligned} L(x, \lambda) &= F(x) + \sum_{i=1}^m \lambda_i G_i(x) \\ &= \mathbf{E} \left\{ f(x, \theta) + \sum_{i=1}^m \lambda_i g_i(x, \theta) \right\} \\ &= \mathbf{E} l(x, \theta, \lambda). \end{aligned} \quad (18)$$

The following inequalities hold:

$$\begin{aligned} F^* &= \min_{x \in X} \max_{\lambda \geq 0} L(x, \lambda) \\ &\geq \max_{\lambda \geq 0} \min_{x \in X} L(x, \lambda) \\ &\geq \max_{\lambda \geq 0} \mathbf{E} \min_{x \in X} l(x, \theta, \lambda) \\ &= \max_{\lambda \geq 0} \mathbf{E} \varphi(\lambda, \theta), \end{aligned}$$

where the function

$$\varphi(\lambda, \theta) = \min_{x \in X} l(x, \theta, \lambda),$$

is concave in λ (it is supposed to be integrable in θ). Thus, for any $\lambda \geq 0$, the quantity

$$h(\lambda) = \mathbf{E} \varphi(\lambda, \theta),$$

is a dual lower estimate for the optimal value F^* . The quantity

$$h^* = \max_{\lambda \geq 0} \mathbf{E} \varphi(\lambda, \theta),$$

is the optimal dual lower estimate for F^* . The estimates $h(\lambda)$, $\lambda \geq 0$, can be calculated, for example, by Monte Carlo simulation and h^* can be found by convex stochastic programming methods (see Ermoliev and Wets 1988). In the last case, one need not solve the stochastic optimization subproblem to optimality: it is possible to stop at any feasible approximation.

When calculating dual estimates, one has to solve the following internal minimization problems

$$\min_{x \in X} l(x, \theta, \lambda),$$

for fixed θ and λ . In many cases these problems can be analytically or numerically solved by nonlinear or discrete programming methods. For example, if functions $f(x, \theta)$ and $g_i(x, \theta)$, $i = 1, \dots, m$, are concave in x and X is a convex polyhedron, then the minimal value of $l(x, \theta, \lambda)$ is achieved at a vertex of X .

3.3. Dual Estimates for Two-stage Stochastic Problems

Let us consider a two-stage stochastic programming problem of the following form:

$$\min_{x \in X} \left[F(x) = \mathbf{E} \left\{ f_1(x, \theta) + \min_{y \in Y(x, \theta)} \sum_{k=1}^K q_k y_k \right\} \right], \quad (19)$$

where

$$Y(x, \theta) = \left\{ y \in R^K : g_i(x, \theta) - \sum_{k=1}^K b_{ik} y_k \leq 0, \right.$$

$$\left. i = 1, \dots, m, y_k \geq 0, k = 1, \dots, K \right\}. \quad (20)$$

Assume that the coefficients q_k and b_{ik} are nonnegative and do not depend on θ , $\sum_{k=1}^K b_{ik} > 0$; functions $f(x, \theta)$ and $g_i(x, \theta)$ are lower semi-continuous in x , integrable in θ , and locally bounded from below by a function integrable in θ . Here, variable x is the first-stage decision, variable y is the second-stage decision (or correction), and θ (environment state) is an elementary event in some probability space $(\Theta, \Sigma, \mathbf{P})$. In (19)–(20) the first-stage decision x has to be made before observing θ , while the second-stage decision y is made when both x and θ are known.

Continuous two-stage stochastic programming problems were extensively studied (see Ermoliev and Wets 1988, Kall 1976, Ermoliev 1976, and Frauendorfer 1992). Our discussion aims at lower bounds for discrete two-stage stochastic programming problems.

It can easily be shown that the function $F(x)$ is lower semi-continuous; hence, the problem has optimal solutions. Denote its optimal value by F^* and define the Lagrangian function:

$$\begin{aligned} L(x, y, \theta, \lambda) &= f_1(x, \theta) + \sum_{k=1}^K q_k y_k \\ &\quad + \sum_{i=1}^m \lambda_i \left(g_i(x, \theta) - \sum_{k=1}^K b_{ik} y_k \right) \\ &= f_1(x, \theta) + \sum_{i=1}^m \lambda_i g_i(x, \theta) \\ &\quad + \sum_{k=1}^K \left(q_k - \sum_{i=1}^m \lambda_i b_{ik} \right) y_k. \end{aligned} \quad (21)$$

The following inequalities are true:

$$\begin{aligned} F^* &= \min_{x \in X} \mathbf{E} \min_{y \geq 0} \max_{\lambda \geq 0} L(x, y, \theta, \lambda) \\ &\geq \min_{x \in X} \mathbf{E} \max_{\lambda \geq 0} \min_{y \geq 0} L(x, y, \theta, \lambda) \\ &= \min_{x \in X} \mathbf{E} \max_{\lambda \in \Lambda} \left\{ f_1(x, \theta) + \sum_{i=1}^m \lambda_i g_i(x, \theta) \right\}, \end{aligned} \quad (22)$$

where

$$\Lambda = \left\{ \lambda \in R^m : \sum_{i=1}^m \lambda_i b_{ik} \leq q_k, \lambda_i \geq 0, i = 1, \dots, m \right\}.$$

Interchanging the operators $\mathbf{E}$ and “min” in (22), we arrive at the estimates:

$$\begin{aligned} F^* &\geq \mathbf{E} \min_{x \in X} \max_{\lambda \in \Lambda} \left\{ f_1(x, \theta) + \sum_{i=1}^m \lambda_i g_i(x, \theta) \right\} \\ &\geq \mathbf{E} \max_{\lambda \in \Lambda} \min_{x \in X} \left\{ f_1(x, \theta) + \sum_{i=1}^m \lambda_i g_i(x, \theta) \right\}. \end{aligned}$$

In this way, we obtained a dual lower bound:

$$L = \mathbf{E} l(\theta),$$

where

$$l(\theta) = \max_{\lambda \in \Lambda} \min_{x \in X} \left\{ f_1(x, \theta) + \sum_{i=1}^m \lambda_i g_i(x, \theta) \right\}.$$

The stochastic bound $l(\theta)$ can be obtained by solving a convex max-min problem. Note again that one need not solve the estimate problem to optimality and can stop at any feasible approximation.

In many cases, the bound $l(\theta)$ is easy to calculate. An important class of such problems are linear problems with

$$f_1(x, \theta) = \sum_{j=1}^n c_j(\theta) x_j,$$

$$g_i(x, \theta) = d_i(\theta) - \sum_{j=1}^n a_{ij}(\theta) x_j, \quad i = 1, \dots, m.$$

(We already assumed the linearity of the second-stage problem in (19)–(20).) Then the stochastic estimate $l(\theta)$ becomes particularly simple:

$$l(\theta) = \max_{\lambda \in \Lambda} \left\{ \sum_{i=1}^m d_i(\theta) \lambda_i + \min_{x \in X} \sum_{j=1}^n \left(c_j(\theta) - \sum_{i=1}^m \lambda_i a_{ij}(\theta) \right) x_j \right\}. \quad (23)$$

Let, additionally, $X = X_0 \cap K$ be an intersection of a set X_0 given by deterministic constraints, and of a set

$$K = \{x \in R^n : \alpha_j \leq x_j \leq \beta_j, x_j \text{ integer}, j = 1, \dots, n\},$$

where α_j and β_j , $j = 1, \dots, n$, are integers. Assume that nonemptiness of X can be easily checked (cf. the Deletion Step of the branch and bound algorithm). Then the estimate (23) simplifies by replacing X by K . The minimization with respect to $x \in K$ has a closed-form solution

$$x_j^*(\lambda, \theta) = \begin{cases} \alpha_j & \text{if } c_j(\theta) - \sum_{i=1}^m \lambda_i a_{ij}(\theta) \geq 0, \\ \beta_j & \text{if } c_j(\theta) - \sum_{i=1}^m \lambda_i a_{ij}(\theta) < 0, \end{cases}$$

which can be substituted into (23):

$$l(\theta) = \max_{\lambda \in \Lambda} \left\{ \sum_{i=1}^m d_i(\theta) \lambda_i + \sum_{j=1}^n \min \left[\left(c_j(\theta) - \sum_{i=1}^m \lambda_i a_{ij}(\theta) \right) \alpha_j, \left(c_j(\theta) - \sum_{i=1}^m \lambda_i a_{ij}(\theta) \right) \beta_j \right] \right\}. \quad (24)$$

It is also possible to derive deterministic dual lower bounds, which, however, do not equal the true value of the objective function for singletons (see Assumption 1).

Interchanging the operators $\mathbf{E}$ and “max” in (22) yields:

$$\begin{aligned} F^* &\geq \min_{x \in X} \max_{\lambda \in \Lambda} \left\{ \mathbf{E} f_1(x, \theta) + \sum_{i=1}^m \lambda_i \mathbf{E} g_i(x, \theta) \right\} \\ &\geq \max_{\lambda \in \Lambda} \min_{x \in X} \left\{ \mathbf{E} f_1(x, \theta) + \sum_{i=1}^m \lambda_i \mathbf{E} g_i(x, \theta) \right\} \\ &\geq \max_{\lambda \in \Lambda} \mathbf{E} \min_{x \in X} \left\{ f_1(x, \theta) + \sum_{i=1}^m \lambda_i g_i(x, \theta) \right\}. \end{aligned}$$

In this way, two deterministic dual lower bounds have been obtained:

$$L_1 = \max_{\lambda \in \Lambda} \min_{x \in X} \left\{ \mathbf{E} f_1(x, \theta) + \sum_{i=1}^m \lambda_i \mathbf{E} g_i(x, \theta) \right\},$$

and

$$L_2 = \max_{\lambda \in \Lambda} \mathbf{E} \min_{x \in X} \left\{ f_1(x, \theta) + \sum_{i=1}^m \lambda_i g_i(x, \theta) \right\}.$$

Estimates L_1 and L_2 are obtained by solving convex stochastic programming problems. For solution techniques for such problems, see Nurminski (1979) and Ermoliev and Wets (1988). As before, any feasible approximation is sufficient.

In the linear case, similarly to (24) we obtain

$$L_1 = \max_{\lambda \in \Lambda} \left\{ \sum_{i=1}^m \mathbf{E} d_i(\theta) \lambda_i + \sum_{j=1}^n \min \left[\left(\mathbf{E} c_j(\theta) - \sum_{i=1}^m \lambda_i \mathbf{E} a_{ij}(\theta) \right) \alpha_j, \left(\mathbf{E} c_j(\theta) - \sum_{i=1}^m \lambda_i \mathbf{E} a_{ij}(\theta) \right) \beta_j \right] \right\}.$$

As an illustration, let us consider the facility location problem of section 1.

Example 7. Facility location (continued). There are two groups of constraints in the second-stage problem: equality constraints

$$d_i(\theta) - \sum_{j=0}^n y_{ij} = 0, \quad i = 1, \dots, m,$$

and inequalities

$$-u_j x_j + \sum_{i=1}^m y_{ij} \leq 0, \quad j = 1, \dots, n.$$

We shall denote by μ_i , $i = 1, \dots, m$, Lagrange multipliers associated with the equalities and by λ_j , $j = 1, \dots, n$, the multipliers associated with the inequalities.

Elementary calculations lead to the following form of the stochastic lower bound (23):

$$l(\theta) = \max_{(\mu, \lambda) \in \Lambda} \left[\sum_{i=1}^m d_i(\theta) \mu_i + \min_{x \in X} \sum_{j=1}^n (c_j - \lambda_j u_j) x_j \right],$$

where

$$\begin{aligned} \Lambda = \{(\mu, \lambda) \in R^{m \times n} : &\mu_i - \lambda_j \leq q_{ij}, \\ &i = 1, \dots, m, j = 1, \dots, n, \\ &\lambda_j \geq 0, j = 1, \dots, n\}, \end{aligned}$$

and

$$X = \{0, 1\}^n.$$

Minimization in x and maximization in μ (for $d(\theta) \geq 0$) can be carried out analytically:

$$x_j^*(\lambda_j) = \begin{cases} 0 & \text{if } c_j \geq \lambda_j u_j, \\ 1 & \text{if } c_j \leq \lambda_j u_j; \end{cases}$$

$$\mu_i^*(\lambda) = \min_{1 \leq j \leq n} (q_{ij} + \lambda_j).$$

Finally, we obtain the following stochastic lower bound:

$$l(\theta) = \max_{\lambda \geq 0} \left[\sum_{i=1}^m d_i(\theta) \min_{1 \leq j \leq n} (q_{ij} + \lambda_j) + \sum_{j=1}^n \min(0, c_j - \lambda_j u_j) \right],$$

which can be easily calculated by standard linear programming techniques. By replacing $d_i(\theta)$ by the expected values $\mathbf{E}d_i(\theta)$, a deterministic lower bound can be obtained.

It is worth noting that for two-stage stochastic programming problems with integer recourse considered in Laporte and Louveaux (1993), Louveaux and van der Vlerk (1993), and Van der Vlerk (1995), similar lower bounds in closed form can also be derived.

If the definition of the set X in (19) involves some constraints of form (16), it is possible to combine the techniques of the last two sections to derive dual bounds for such a problem. Technically, it means augmenting the Lagrangian function (21) with terms associated with direct constraints (16), as in (18). All the remaining steps are essentially the same as above.

4. ILLUSTRATIVE EXAMPLES

Let us discuss some advantages of the proposed approach in contrast to the conventional optimization techniques. For this purpose we consider a simplified version of the project financing problem (Examples 3 and 6) for the case of finitely many possible outcomes (scenarios) $\theta^1, \dots, \theta^S$. We also assume that there is only one deterministic resource. Under these assumptions we obtain an equivalent large scale 0–1 programming problem:

$$\min \left[\sum_{j=1}^n c_j x_j - \frac{1}{S} \sum_{s=1}^S \sum_{j=1}^n q_j^s y_j^s \right], \quad (25)$$

$$\sum_{j=1}^n d_j x_j + \sum_{j=1}^n \delta_j^s y_j^s \leq b, \quad s = 1, \dots, S, \quad (26)$$

$$\{0, 1\} \ni y_j^s \leq x_j \in \{0, 1\}, \quad (27)$$

$$j = 1, \dots, n, s = 1, \dots, S.$$

Here $q_j^s = q_j(\theta^s)$, $\delta_j^s = \delta_{ij}(\theta^s)$, and the subscript i is dropped from the resource constraints. Still, with all these simplifications (25)–(27) is a rather large problem with $(S + 1)n$ binary variables, S knapsack constraints (26) and Sn implications (27).

The advantage of our approach is that instead of the optimization problem (25)–(27) in the very large space of variables x and y , we deal with the optimization of an implicitly given function $F(x)$ of variables x , defined by (3). Observe that for a given x , the calculation of the minimum value of (25)–(27) with respect to y decomposes into S independent knapsack problems, which can be

quickly solved by dynamic programming (see Nemhauser and Wolsey 1988, p. 420–421).

To generate upper and lower bounds for this problem we use the techniques discussed in Section 3.

Upper Bound. To obtain an upper bound for some set $X' = \{x : x_j \in \{l_j, u_j\}, j = 1, \dots, n\}$ we consider at first the expected value problem:

$$\min \sum_{j=1}^n (c_j - \mathbf{E}q_j(\theta))x_j,$$

$$\sum_{j=1}^n (d_j + \mathbf{E}\delta_j(\theta))x_j \leq b, \quad i = 1, \dots, n,$$

$$x \in X'.$$

An upper bound solution can be generated by the following simple greedy algorithm. Let π be a permutation of $\{1, \dots, n\}$ in the order of nondecreasing values of the ratios

$$r_j = \frac{c_j - \mathbf{E}q_j(\theta)}{d_j + \mathbf{E}\delta_j(\theta)}, \quad j = 1, \dots, n, \quad (28)$$

so $r_{\pi(t+1)} \geq r_{\pi(t)}$, $t = 1, \dots, n$. For all j such that $l_j = u_j$ we set $x_j' = l_j$ and we correct the resource by calculating $b' = b - \sum_{j=1}^n l_j d_j$. Finally, we allocate the remaining resource to other projects in the order given by π (Nemhauser and Wolsey 1988, p. 452).

For the point x' obtained in this way the objective value is calculated by solving the second stage knapsack problem for all S scenarios. By using randomly sampled scenarios one obtains a stochastic upper bound.

Lower Bound. To obtain a lower bound for $X' = \{x : x_j \in \{l_j, u_j\}, j = 1, \dots, n\}$ we use the idea of interchanging minimization and expectation operators (9), which leads to the following procedure. For some chosen cost corrections $\lambda_j^s \geq -c_j$, such that $\sum_{s=1}^S \lambda_j^s = 0$, $j = 1, \dots, n$, we solve by dynamic programming for $s = 1, \dots, S$ the knapsack subproblems

$$\min \sum_{j=1}^n ((1 - l_j)(c_j + \lambda_j^s) - q_j^s) y_j^s, \quad (29)$$

$$\sum_{j=1}^n ((1 - l_j)d_j + \delta_j^s) y_j^s \leq b - \sum_{j=1}^n l_j d_j, \quad (30)$$

$$y_j \in \{0, u_j\}, \quad j = 1, \dots, n. \quad (31)$$

Denoting the optimal values of these problems by $\varphi^s(X')$, $s = 1, \dots, S$ one obtains the following lower bound

$$L(X) = \sum_{s=1}^S l_j c_j + \frac{1}{S} \sum_{s=1}^S \varphi^s(X').$$

Two cases of the cost corrections were considered.

Case 1. No correction. In this case we simply set $\lambda_j^s = 0$ for $s = 1, \dots, S$ and $j = 1, \dots, n$. This corresponds to the basic bound (9).

Case 2. Efficiency correction. To motivate this correction let us note that the quality of the lower-bound generated

Table I
Performance of the Branch-and-Bound Methods

Problem			Stochastic Branch-and-Bound Method				CPLEX	
			No correction		Cost correction			
Projects	Scenarios	Distribution	Nodes	Time	Nodes	Time	Nodes	Time
10	10	u	30	2.38	9	1.02	7	0.18
20	10	u	52	4.22	13	2.04	223	5.25
50	10	u	193	42.06	80	18.95	1,018	116.05
100	10	u	6,736	2,577.85	557	208.59	8,896	3,355.37
10	10	n-u	10	0.41	10	0.41	1,295	16.13
20	10	n-u	19	1.52	19	1.53	1,776	40.14
50	10	n-u	3,947	304.05	3,463	285.06		Failed
100	10	n-u	23,647	3,203.84	22,891	3,115.44		Failed
10	100	u	27	19.95	12	9.25		Failed
20	100	u	63	50.72	23	22.48		Failed
50	100	u	372	737.73	133	276.74		Failed
100	100	u	7,809	27,546.30	1,870	6,483.71		Failed
10	100	n-u	15	5.05	15	5.06		Failed
20	100	n-u	30	20.27	28	19.06		Failed
50	100	n-u	4,345	3,114.74	4,076	2,956.50		Failed
100	100	n-u	17,432	22,657.30	16,937	22,298.43		Failed

“u” and “n-u” denote the uniform and nonuniform distribution.

“Nodes” denotes the number of subsets generated.

“Time” is the CPU time in seconds on a SUN SPARCstation 2.

“Failed” means that CPLEX reached the maximum number of nodes (10^5).

in the basic case depends on the differences among the first-stage decisions $\hat{x}_j^s = \max\{l_j, \hat{y}_j^s\}$, which follow from the solutions $\hat{y}^s$ of (29)–(31) for $s = 1, \dots, S$. To make the scenario subproblems as similar as possible, λ_j^s are chosen in such a way that the relative efficiency of project j ,

$$r_j^s = \frac{c_j - \lambda_j^s - q_j^s}{d_j + \delta_j^s},$$

is the same for all scenarios s . Equating r_j^s to the efficiency (28) in the expected value problem we obtain:

$$\bar{\lambda}_j^s = c_j - q_j^s - (d_j + \delta_j^s) \frac{c_j - \mathbf{E}q_j(\theta)}{d_j + \mathbf{E}\delta_j(\theta)}.$$

Observe that by construction $\sum_{s=1}^S \bar{\lambda}_j^s = 0$ for all j . To make sure that $c + \lambda^s \geq 0$ for all s , we define $\lambda_j^s = \sigma_j \bar{\lambda}_j^s$, where $\sigma_j \in [0, 1]$, $j = 1, \dots, n$, is a scaling factor.

Branching. The branching rule was based on the upper bound solution x' in the record subset: the first (in the sense of the ordering π) nonzero of x' , for which branching was possible, was selected as the next branching variable.

A number of instances of problem (25)–(27) were generated randomly as follows. For a fixed number of projects n the first stage costs and the first stage resource demands were generated randomly from the uniform distribution in the interval $[0, 50]$. The left ends of the intervals for second stage demands and profits, and the lengths of these intervals, were generated in the same way. Then, for a given number of scenarios S , profits and resource demands were generated from these intervals according to the uniform distribution.

Next, a second series of problems were generated. The same procedure was used as before, but the randomly generated numbers were squared. The lower bounds for demands and profits/costs were generated with the use of the same random number, so correlation of these quantities occurred. In this way, a nonuniform distribution of dependent variables was created.

The number of projects ranged through $n = 10, 20, 50, 100$, and two numbers of scenarios were considered: $S = 10$ and $S = 100$. Note that the case $n = 100$ and $S = 100$ correspond to a 0–1 linear programming problem with 10,100 variables, 100 knapsack constraints and 10,000 implications. The resource available was always 500, and there were always 5 to 15 projects started at the optimal solution.

As a benchmark we used the general-purpose MIP programming system CPLEX (1993), which was applied directly to the large-scale problem (25)–(27) with the priority order preferring the first stage variables x , and other parameters set at their default values.

The results are summarized in Table I. We see that the proposed method solved all instances of the problem in a relatively short time. The price correction scheme based on efficiency equilibration was very successful; it always decreased the solution time. The problems with uniform and independent distributions were easier, because the upper bound solution reached the optimal point very quickly. The problems with nonuniform distributions, which had large differences in sizes and profits of projects, turned out to be more difficult for the stochastic method and too difficult for CPLEX. The reason for it is rather clear: our approach has a linear growth of complexity, when the

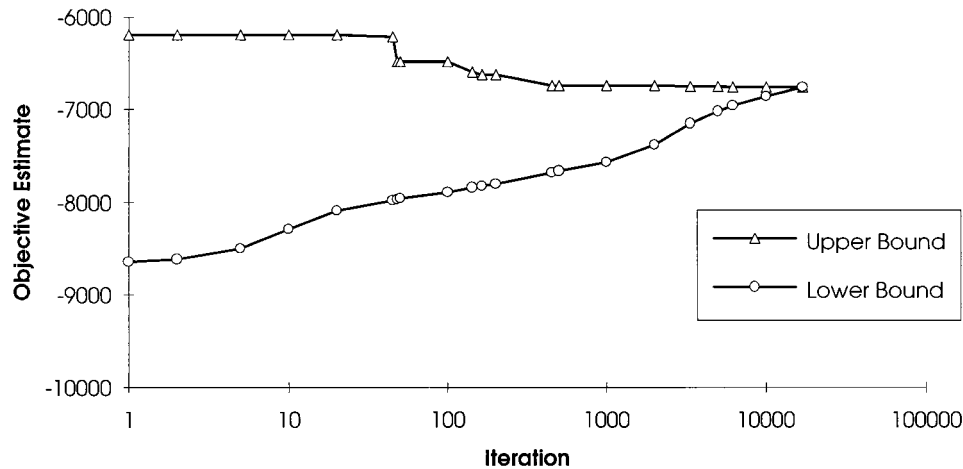


Figure 1. Operation of the method with cost correction on the example with 100 projects and 100 scenarios drawn from a nonuniform distribution.

number of scenarios increases, while in the general branch-and-bound method the growth is exponential. The relatively smaller gain from the cost correction in our second method was due to the fact that small scaling factors σ_j had to be used to ensure the inequalities $c_j + \lambda_j^s \geq 0$.

To illustrate the progress of the method with price correction we show in Figure 1 the best upper and the best lower bound as functions of the iteration number.

Our next experiment aimed at assessing the capabilities of the method for larger problems, where stochastic bounds had to be used. A problem with $n = 10$ and $S = 1000$ and with a nonuniform distribution was generated, and two versions of the method were compared. The first one used deterministic bounds based on all 1000 scenarios (as in the earlier experiments). The second version used stochastic bounds calculated as follows: at each iteration 10 scenarios were drawn at random from the whole population and for these scenarios the bounds were calculated by the techniques described earlier in this section. The stochastic method stopped the first time when the record set was a singleton and the true objective value was calculated on the basis of the whole set of scenarios. Ten such passes (regenerative cycles) of the stochastic method were made, with different seeds for the random number generator used to sample the scenarios. The results are shown in Table II. Additionally, in Figure 2 we illustrate the progress of the deterministic method and the stochastic method in Pass 9 (note the nonmonotonicity of the bounds).

We see that the stochastic method does surprisingly well on these examples, even with such a small sample size. All cycles found solutions that were better than the first heuristic solution based on the expected value problem (the first upper bound in Pass 0), and the optimal solution was hit twice.

The final experiment aimed at assessing the capabilities of the stochastic method on very large problems, which seem to be intractable otherwise. A problem with $n = 100$

projects and with $S = 1000$ nonuniformly distributed scenarios was considered, which had a very large deterministic equivalent. To obtain stochastic bounds, 100 scenarios were drawn at random at each iteration and the lower and upper bounds were calculated as described earlier in this section. The algorithm stopped when the record set was a singleton and this point was considered as a candidate for the solution. Figure 3 illustrates one pass of the method, which took 5734 iterations in 8838 CPU seconds. Of course, we do not know the optimal solution in this case, but the objective value at the solution found (calculated with the use of all scenarios) was approximately 7 percent better than the objective value for the heuristic solution based on the expected value problem.

Table II
Performance of the Stochastic Branch-and-Bound Method on a Problem with 10 Projects and 1,000 Nonuniformly Distributed Scenarios

Pass	Solution	Nodes	Time	Objective
0	0011001110	16	51.31	-1,231.85
1	0000001110	13	1.77	-908.12
2	0001001110	11	2.17	-1,215.66
3	0011011100	13	1.70	-1,011.34
4	1011000110	13	2.01	-962.27
5	0011011110	13	1.96	-1,131.64
6	0011000110	12	2.15	-1,025.69
7	0011001100	19	2.08	-1,080.85
8	0001011000	15	1.64	-835.07
9	0011001110	12	2.16	-1,231.85
10	0011001110	16	2.32	-1,231.85

At Pass 0 exact bounds were calculated.

At Passes 1–10 random bounds based on 10 observations were used.

“Nodes” denotes the number of subsets generated.

“Time” is the CPU time in seconds on a SUN SPARCstation 2, including the time of the final objective estimation.

“Objective” is the true objective value at the solution.

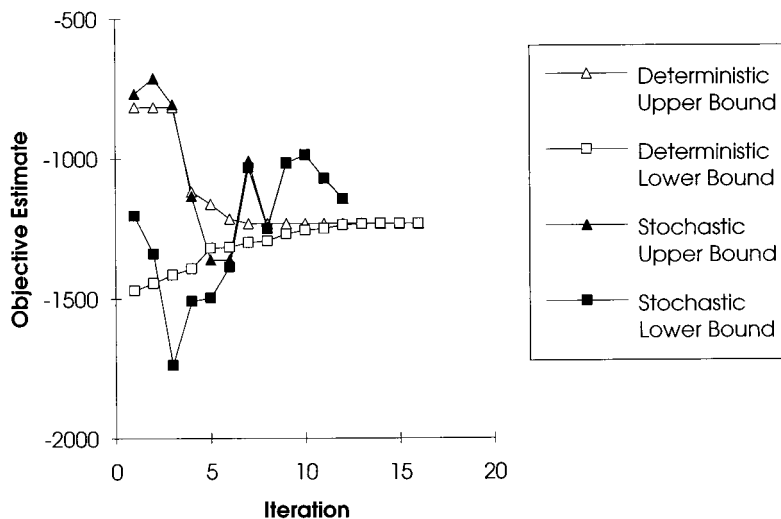


Figure 2. Operation of the methods with deterministic and stochastic bounds on the example with 10 projects and 1000 scenarios drawn from a nonuniform distribution. The stochastic method uses only 10 observations per iteration.

Of course, it is too early to draw definite conclusions about the capabilities of the proposed approach, but it seems to have a potential to solve some large scale stochastic integer programming problems. It appears to be useful (at least in the examples discussed here) to take advantage of the stochastic nature of the problem in the derivation of the bounds and in the partitioning strategy.

5. CONCLUSIONS

The stochastic branch-and-bound method presented in this paper combines two basic ideas. The first one is to partition the set of decisions into smaller subsets and to use bounds on the objective within the subsets to guide this process, similarly to deterministic discrete optimization. The second idea is to exploit the stochastic nature of the problem in order to obtain stochastic or specialized deterministic bounds. We use the concept of recursive allocation of random observations to the subsets to improve

stochastic bounds. As a result, a rather general and flexible scheme has been obtained in which partition and observation can be dependent on the outcomes of the previous observations. The method is convergent with probability one under quite general assumptions. Constructive methods for calculating stochastic bounds for a broad class of problems have been developed. If the problem has a finite discrete distribution of random parameters, then the proposed bounds can be calculated exactly. In this way one obtains a highly specialized deterministic branch and bound method for stochastic programming problems.

Some initial computational experiments with the method indicate that it has a potential to solve some large stochastic discrete optimization problems. However, there are still a number of theoretical and practical questions that have to be investigated.

There is a need to develop a probabilistic concept of efficiency of the method. This would allow to introduce

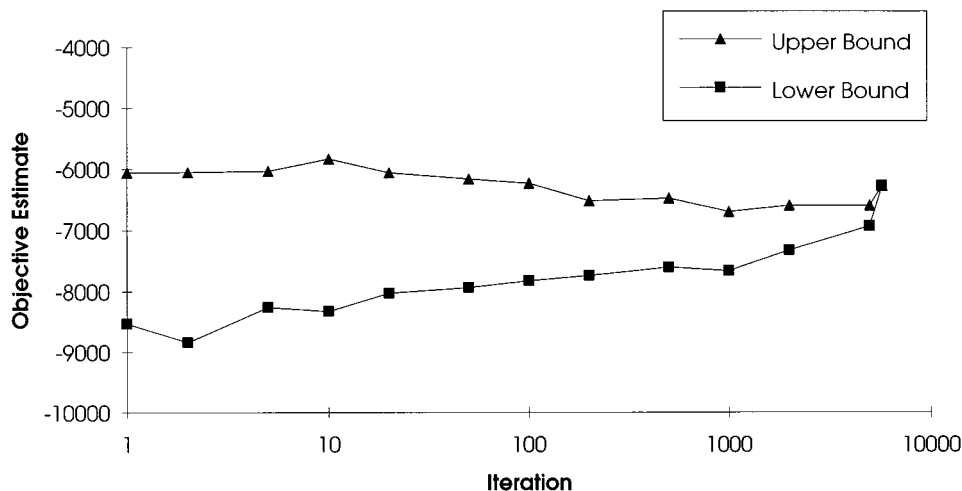


Figure 3. Operation of the stochastic branch-and-bound method on the example with 100 projects and 1000 scenarios drawn from a nonuniform distribution. The stochastic method uses 100 observations per iteration.

more specific partitioning and observation rules. Ideally, one would like to have something similar to the optimal allocation indices of Gittins for the multiarmed bandit problem (Gittins 1989). Indeed, the proposed approach can be viewed as the generalization of the solution procedure for the multiarmed bandit problem with the following difference: observations are made for collections of actions (arms), not just one. Therefore the decision is not only the choice of the next action, but also the partition of the subsets. It is clearly a much more difficult problem, but with a great theoretical and practical importance. Presumably, some more detailed results can be obtained for some specific classes of stochastic discrete problems. Finally, further computational experience has to be gained for a sufficiently broad class of application problems to better understand the advantages and the disadvantages of various possible versions of the method. We hope to make some progress in these directions in the near future.

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